

Product Requirements Document (PRD)

Phase 1: Hospital Management System (HMS) Core Web Application & Admin Panel

Document Version: 1.0

Project Name: Hospital Management System (HMS)

Target Platform: Cloud-based Responsive Web Application

Estimated Timeline: 25 Working Days

1. Product Vision & Scope

The Hospital Management System (HMS) is a centralized ERP platform engineered to digitize and automate operational workflows for healthcare centers, hospitals, and clinics. This phase establishes the core backend architecture, databases, and responsive web panels required to manage day-to-day hospital operations, financial accounting, inventory, and partner commissions.

2. Technical Architecture & Stack

Component	Technology / Platform
Backend Framework	Laravel (Latest Version)
Frontend Framework	Vue.js (Responsive Web Application)
Database	MySQL
API Architecture	REST APIs (Secure & Encrypted)
Push Notifications	Firebase Cloud Messaging
Version Control	GitHub Repository
Infrastructure	Cloud Hosting with SSL configuration

3. User Roles & Access Control (RBAC)

Role-Based Access Control (RBAC) is mandatory to ensure data security and operational isolation.

User Role	Primary Responsibilities & Access Rights
Super Admin	Full system access, configuration, commission setting, and comprehensive reporting.
Receptionist	Patient registration, appointment booking, OPD/IPD intake, and token management.
Doctor	Profile management, schedule availability, prescription history, and leave management.
Accountant	Invoicing, cash/bank book management, ledgers, and profit & loss reporting.
Pharmacist	Medicine inventory, barcode scanning, supplier tracking, and pharmacy billing.
Lab Technician	Sample tracking, test category management, and PDF report generation.
Franchise Partner	Referral dashboard access, commission wallet viewing, and withdrawal requests.

4. Detailed Module Specifications

4.1. Patient & Appointment Management

Generate unique UHID for all new patient registrations.

Store and retrieve comprehensive medical history, previous visits, and uploaded documents.
Facilitate online appointment booking, walk-in registrations, and automated token management.
Trigger SMS reminders for upcoming appointments.

4.2. OPD, IPD & Bed Management

Process OPD registrations and IPD admissions seamlessly.
Record treatment notes, nursing notes, and generate discharge summaries.
Monitor real-time bed availability across ICU, General Ward, Private, and Deluxe Rooms.
Execute bed allocations, bed transfers, and generate occupancy reports.

4.3. Financial Accounting & Billing

Generate consolidated and module-specific bills (OPD, IPD, Pharmacy, Laboratory, Packages).
Produce GST-compliant invoices and facilitate receipt printing.
Track income and expenses through a digital cash book and bank book.
Maintain accurate ledgers, balance sheets, and profit & loss statements.

4.4. Pharmacy & Laboratory

Manage medicine inventory, purchase orders, and supplier relationships.
Implement stock alerts for low inventory and expiry tracking.
Enable barcode support for rapid pharmacy billing.
Configure test categories, manage sample collections, and automate PDF test report generation.

4.5. Franchise / Channel Partner Ecosystem

Provide a dedicated registration and dashboard interface for diagnostic centers, NGOs, and consultants.
Track patient referrals and automatically calculate commissions based on Admin-set parameters (Fixed, Doctor-wise, Service-wise, Surgery-wise).
Manage a commission wallet, withdrawal requests, and referral analytics.

5. Non-Functional & Security Requirements

Data Protection	Implement an encrypted MySQL database with secure REST APIs.
Infrastructure Security	Mandate SSL support across all web interfaces.
Data Integrity	Configure automated daily backup support.
Traceability	Maintain comprehensive audit logs for all critical system actions.